PSBT03 BTHome Low Power Human Presence Sensor Application Manual v1.1

1.1 UART Interface

The serial port uses TTL-level communication. The protocol uses AT-like commands. Each command ends with 0x0D 0x0A (\r\n).

1.1.1 Communication Test Command

AT Test Command	
Execute command	AT
Normal response	OK
Error response	AT+ERR

1.1.2 Baud Rate

AT+BAUD Baud Rate	
Set command	AT+BAUD=<num>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+BAUD?
Query response	AT+BAUD=<num> If there is an error, returns AT+ERR.

<num>: baud rate index Saved after power cycle	Valid range: 1 <= num <= 8 1: 9600 2: 19200 3: 38400 4: 57600 5: 115200 (default) 6: 230400 7: 256000 8: 460800

1.1.3 GPIO Output Level Configuration

AT+GPIO GPIO Output Level Configuration	
Set command	AT+GPIO=<level>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+GPIO?
Query response	AT+GPIO=<level> If there is an error, returns AT+ERR.
<level>: output level setting Saved after power cycle	0 (default): output low when no target is triggered; output high when a target is triggered. 1: output high when no target is triggered; output low when a target is triggered.

1.1.4 Target Output Enable

AT+TAGOUT Target Output Enable	
Set command	AT+TAGOUT=<targetout_en>
Set response	AT+OK If there is an error, returns AT+ERR.

Query command	AT+TAGOUT?
Query response	AT+TAGOUT=<targetout_en> If there is an error, returns AT+ERR.
<targetout_en>: target output enable Saved after power cycle	0: the serial port outputs nothing. 1: outputs only the target-absent / target-present state (0 or 1). 2 (default): outputs the target-absent / target-present state (0 or 1) together with the corresponding target information, including distance and energy.

The default serial output content is interpreted as follows:

Default Serial Output Example	
0	No target detected
1	Target detected
Distance	Target distance, in cm
Energy	Target energy value

1.1.5 Hold Frame Count

AT+HOLD Hold Frame Count	
Set command	AT+HOLD=<holdperiod>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+HOLD?
Query response	AT+HOLD=<holdperiod> If there is an error, returns AT+ERR.
<holdperiod>: hold frame count	Valid range: 1 <= holdperiod <= 100000

Saved after power cycle	If no target is detected for holdperiod consecutive frames while the sensor is in the occupied state, the state changes to unoccupied. Default: 40.

1.1.6 Target Detection Distance

Target detection is divided into three distance zones. The distance breakpoints are R1, R2, and R3, with increasing values, that is, 0 <= R1 <= R2 <= R3.

AT+R1 Target Detection Distance 1	
Set command	AT+R1=<r1>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+R1?
Query response	AT+R1=<r1> If there is an error, returns AT+ERR.
<r1>: target detection distance 1 Saved after power cycle	Valid range: 0 <= r1 <= 2000 Default: 200 cm.

AT+R2 Target Detection Distance 2	
Set command	AT+R2=<r2>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+R2?
Query response	AT+R2=<r2> If there is an error, returns AT+ERR.
<r2>: target detection distance 2 Saved after power cycle	Valid range: 0 <= r2 <= 2000 Default: 500 cm.

AT+R3 Target Detection Distance 3	
Set command	AT+R3=<r3>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+R3?
Query response	AT+R3=<r3> If there is an error, returns AT+ERR.
<r3>: target detection distance 3 Saved after power cycle	Valid range: 0 <= r3 <= 2000 Default: 1000 cm.

1.1.7 Target Trigger Threshold

The trigger threshold for each distance zone can be set independently as MR1TH, MR2TH, and MR3TH. MR1TH applies to targets within [0, R1], MR2TH applies to targets within (R1, R2], and MR3TH applies to targets within (R2, R3].

AT+MR1TH Distance 1 Target Trigger Threshold	
Set command	AT+MR1TH=<mr1th>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+MR1TH?
Query response	AT+MR1TH=<mr1th> If there is an error, returns AT+ERR.
<mr1th>: distance 1 target trigger threshold Saved after power cycle	Valid range: 1 <= mr1th <= 10000 Default: 14 A smaller value provides higher sensitivity.

AT+MR2TH Distance 2 Target Trigger Threshold	
Set command	AT+MR2TH=<mr2th>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+MR2TH?
Query response	AT+MR2TH=<mr2th> If there is an error, returns AT+ERR.
<mr2th>: distance 2 target trigger threshold Saved after power cycle	Valid range: 1 <= mr2th <= 10000 Default: 10 A smaller value provides higher sensitivity.

AT+MR3TH Distance 3 Target Trigger Threshold	
Set command	AT+MR3TH=<mr3th>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+MR3TH?
Query response	AT+MR3TH=<mr3th> If there is an error, returns AT+ERR.
<mr3th>: distance 3 target trigger threshold Saved after power cycle	Valid range: 1 <= mr3th <= 10000 Default: 10 A smaller value provides higher sensitivity.

1.1.8 Target Hold Threshold

The hold threshold for each distance zone can be set independently as R1TH, R2TH, and R3TH. R1TH applies to targets within [0, R1], R2TH applies to targets within (R1, R2], and R3TH applies to targets within (R2, R3].

AT+R1TH Distance 1 Target Hold Threshold

Set command	AT+R1TH=<r1th>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+R1TH?
Query response	AT+R1TH=<r1th> If there is an error, returns AT+ERR.
<r1th>: distance 1 target hold threshold Saved after power cycle	Valid range: 1 <= r1th <= 10000 Default: 10 A smaller value provides higher sensitivity.

AT+R2TH Distance 2 Target Hold Threshold	
Set command	AT+R2TH=<r2th>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+R2TH?
Query response	AT+R2TH=<r2th> If there is an error, returns AT+ERR.
<r2th>: distance 2 target hold threshold Saved after power cycle	Valid range: 1 <= r2th <= 10000 Default: 6 A smaller value provides higher sensitivity.

AT+R3TH Distance 3 Target Hold Threshold	
Set command	AT+R3TH=<r3th>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+R3TH?

Query response	AT+R3TH=<r3th> If there is an error, returns AT+ERR.
<r3th>: distance 3 target hold threshold **Saved after power cycle**	Valid range: 1 <= r3th <= 10000 Default: 6 A smaller value provides higher sensitivity.

The relationship between the target detection range and the corresponding trigger and hold thresholds is shown below.

Target Detection Range	Corresponding Trigger Threshold	Corresponding Hold Threshold
0 to R1	MR1TH	R1TH
R1 to R2	MR2TH	R2TH
R2 to R3	MR3TH	R3TH

1.1.9 Trigger Sensitivity

AT+TRITH Trigger Sensitivity	
Set command	AT+TRITH=<level>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+TRITH?
Query response	AT+TRITH=<level> If there is an error, returns AT+ERR.
<level>: trigger sensitivity **Saved after power cycle**	Valid range: 1 <= level <= 10 While the sensor is in the unoccupied state, if a target is detected in level frames out of ten consecutive frames, the state switches to occupied. Default: 3.

1.1.10 State Hold Threshold

AT+ONTH Occupied-State Hold Threshold	
Set command	AT+ONTH=<holdonth>
Set response	AT+OK If there is an error, returns AT+ERR.
Query command	AT+ONTH?
Query response	AT+ONTH=<holdonth> If there is an error, returns AT+ERR.
<holdonth>: state hold threshold Saved after power cycle	Valid range: 1 <= holdonth <= 10 After the state changes to occupied, if a target is detected in holdonth frames out of ten consecutive frames, the occupied state is confirmed. Default: 3.

1.1.11 State Assignment

AT+OUTTRI State Assignment	
Set command	AT+OUTTRI=<num>
Set response	AT+OK If there is an error, returns AT+ERR.
<num>: state parameter	num=1: assigns the radar sensor state to indicate that the sensor has switched to the occupied state. Used in multi-sensor fusion scenarios.

1.1.12 Operating Frequency

AT+FREQ Operating Frequency	
Set command	AT+FREQ=<num>
Set response	AT+OK If there is an error, returns AT+ERR.

<num>: operating frequency parameter Saved after power cycle	Valid range: 0 <= num <= 4 0: 0.5 Hz, operating period 2 s 1: 1 Hz, operating period 1 s 2: 2 Hz, operating period 0.5 s 3: 4 Hz, operating period 0.25 s 4: 8 Hz, operating period 0.125 s This parameter provides a convenient way to change the operating frequency of the radar sensor. Default: 2, so the default operating period is 0.5 s.

1.1.13 Operating Period

If the preset operating frequencies do not meet the application requirements, the operating period can also be configured directly. It is divided into a fast operating period and a slow operating period. The fast operating period is used after a target is detected, while the slow operating period is used when no target is detected.

AT+FTIME Fast Operating Period	
Set command	AT+FTIME=<ftime>
Set response	AT+OK If there is an error, returns AT+ERR.
<ftime>: fast operating period Saved after power cycle	Valid range: 5 <= ftime <= 100000 Default: 500 ms.

AT+STIME Slow Operating Period	
Set command	AT+STIME=<stime>
Set response	AT+OK If there is an error, returns AT+ERR.
<stime>: slow operating period Saved after power cycle	Valid range: 5 <= stime <= 100000 Default: 500 ms.

1.1.14 Restore Factory Defaults

AT+INIT Restore Factory Defaults	
Set command	AT+INIT
Response	AT+OK All parameters are restored to the factory default values, and then the software resets and restarts using the restored parameters.

1.1.15 Software Reset Command

AT+RESET Software Reset Command	
Execute command	AT+RESET
Response	Returns the current parameter values. After parameter configuration is complete, send this command to perform a software reset and restart using the new parameter settings.

1.1.16 Sleep Command

AT+SLEEP Sleep Command	
Execute command	AT+SLEEP
Response	AT+OK The radar sensor enters sleep mode. At this time, the GPIO pin indicates the unoccupied state, and the serial port outputs no data. Wake-up procedure: 1) Send any command or data (able to generate a 500 us low pulse) to wake the radar, then wait for the radar to return STOP.

	2) After the radar returns STOP, send AT+START if you want the sensor to continue running from the state it had before sleep. If you want the radar sensor to reset and restart, send AT+RESET.